# Supplementary Material for "Nodal Frequency-Secure Real-Time Dispatch of Inverter-Dominated Power Systems With Learned Constraints Monotone in Activatable Reserve"

This document details the nodal post-contingency model of Section II-A and the training of the PICNN of Section III-B of the paper. Equations and tables numbered with the prefix S belong to this document, and all other numbers refer to the paper.

## I. Nodal Post-Contingency Model

The model has the structure of the nodal frequency response models used in frequency-constrained scheduling and nodal frequency analysis [1]–[4]. Synchronous generators (SGs) and grid-forming (GFM) inverter-based resources (IBRs) are swing devices with (virtual) inertia, damping, and a power–frequency control loop. Grid-following (GFL) IBRs and battery energy storage system (BESS) units inject their frequency response at their buses. All devices are coupled through the network susceptances, and the bus frequencies follow from the device frequencies by the frequency divider [5]. Every unit is represented individually, and its response is limited by its reserve in the dispatch. Powers are in p.u. on the base $P^B$ unless stated otherwise.

### A. Devices

For contingency $c$, let $\mathcal{S}_c$ be the surviving swing devices, which are the SGs with positive active-power capability and the GFM IBRs online at the penetration level, and let $\mathcal{F}_c$ be the surviving GFL IBRs and BESS units. A GFM IBR is represented, like an SG, by a swing device with its virtual inertia [2]. Swing device $u$ at bus $n(u)$, with rating $S_u$, inertia constant $H_u$ on its rating ($H^{\mathrm{V}}$ for a GFM IBR), and damping $D_u$, obeys

$$\dot{\delta}_u = 2\pi f_0(\omega_u - 1), \quad 2M_u\dot{\omega}_u = p^{\mathrm{m}}_u - p^{\mathrm{e}}_u - D_u(\omega_u - 1), \tag{S1}$$

where $M_u = H_u S_u / P^B$, $\delta_u$ and $\omega_u$ are its internal angle and per-unit speed, and $p^{\mathrm{m}}_u$ and $p^{\mathrm{e}}_u$ are its mechanical or control power and its electrical power. Each swing device follows the droop setpoint

$$g_u = \mathrm{sat}_{[\underline{p}_u, \overline{p}_u]}\Big(P_u + \frac{S_u}{0.05P^B}(1 - \omega_u)\Big), \tag{S2}$$

where $P_u$ and $R_u$ are its output and reserve in the dispatch, $\overline{p}_u = P_u + R_u$, $\underline{p}_u = \max(P_u - R_u, 0)$, and $\mathrm{sat}_{[a,b]}$ clips its argument to $[a, b]$. The droop is 5% on the rating. The governor and turbine of an SG have the structure of the TGOV1 model [6], a valve lag $T_1$ followed by a lead–lag reheat stage $(T_2, T_3)$:

$$\begin{aligned} T_1\dot{x}_{1,u} &= g_u - x_{1,u}, \quad T_3\dot{x}_{2,u} = (1-a)\,x_{1,u} - x_{2,u}, \\ p^{\mathrm{m}}_u &= a\,x_{1,u} + x_{2,u}, \quad a = T_2/T_3. \end{aligned} \tag{S3}$$

A GFM IBR delivers its setpoint through a first-order activation lag [4],

$$T^{\mathrm{A}}\dot{p}^{\mathrm{m}}_u = g_u - p^{\mathrm{m}}_u. \tag{S4}$$

The units in $\mathcal{F}_c$ respond to the center-of-inertia (COI) frequency

$$f^{\mathrm{coi}} = f_0\Big[1 + \sum_{u\in\mathcal{S}_c} M_u(\omega_u - 1)\Big/\sum_{u\in\mathcal{S}_c} M_u\Big] \tag{S5}$$

through a first-order lag,

$$\tau_j\dot{p}^{\mathrm{F}}_j = \mathrm{sat}_{[0,\overline{r}_j]}\Big(\frac{S_j}{0.05P^B}\Big(1 - \frac{f^{\mathrm{coi}}}{f_0}\Big)\Big) - p^{\mathrm{F}}_j, \quad j \in \mathcal{F}_c, \tag{S6}$$

where $p^{\mathrm{F}}_j$ is the incremental injection of unit $j$ and $S_j$ its rating. A GFL IBR $i$ has $\overline{r}_j = R_i$ and $\tau_j = T^{\mathrm{A}}$, and a BESS unit $b$ has $\overline{r}_j = r^{\mathrm{F}}_b/P^B$ and $\tau_j = 0.1$ s.

### B. Network and Bus Frequencies

Let $\boldsymbol{B}^{\mathrm{net}}$ be the susceptance Laplacian of the in-service branches, in which branch $(n, m)$ with reactance $x_{nm}$ adds $1/x_{nm}$ to the entries $(n, n)$ and $(m, m)$ and $-1/x_{nm}$ to the entries $(n, m)$ and $(m, n)$. The internal node of swing device $u$ is coupled to bus $n(u)$ through $x_u = X_u P^B/S_u$, where $X_u$ is the transient reactance of an SG or the coupling reactance of a GFM IBR on its rating. With $\boldsymbol{X} = \mathrm{diag}(x_u)$ and the bus incidence matrices $\boldsymbol{E}_{\mathrm{S}}$ of the swing devices and $\boldsymbol{E}_{\mathrm{F}}$ of the units in $\mathcal{F}_c$, let

$$\boldsymbol{B}_{\mathrm{BB}} = \boldsymbol{B}^{\mathrm{net}} + \boldsymbol{E}_{\mathrm{S}}\boldsymbol{X}^{-1}\boldsymbol{E}_{\mathrm{S}}^{\mathsf{T}}, \quad \boldsymbol{B}_{\mathrm{BS}} = \boldsymbol{E}_{\mathrm{S}}\boldsymbol{X}^{-1}. \tag{S7}$$

The bus angles $\boldsymbol{\theta}$ and the electrical powers of the swing devices follow from

$$\boldsymbol{B}_{\mathrm{BB}}\boldsymbol{\theta} = \boldsymbol{B}_{\mathrm{BS}}\boldsymbol{\delta} + \boldsymbol{p}^{\mathrm{B}} + \boldsymbol{E}_{\mathrm{F}}\boldsymbol{p}^{\mathrm{F}}, \quad p^{\mathrm{e}}_u = (\delta_u - \theta_{n(u)})/x_u, \tag{S8}$$

where $\boldsymbol{p}^{\mathrm{B}}$ collects the scheduled outputs of the GFL IBRs and BESS units minus the bus demands, which are frequency-independent. The bus frequencies follow from the frequency divider [5]:

$$\boldsymbol{f} = f_0\big[\mathbf{1} + \boldsymbol{B}_{\mathrm{BB}}^{-1}\boldsymbol{B}_{\mathrm{BS}}(\boldsymbol{\omega} - \mathbf{1})\big]. \tag{S9}$$

### C. Contingency Simulation and Worst-Bus Drop

The pre-contingency state is the equilibrium of the intact system at the dispatch, with $p^{\mathrm{m}}_u = p^{\mathrm{e}}_u = P_u$ for every swing device and the bus demands scaled by a common factor that allocates the network losses of the dispatch in proportion to them. At $t = 0$ the units of $\mathcal{T}_c$ trip: their internal nodes, scheduled outputs, inertia, and reserve are removed, and $\boldsymbol{B}_{\mathrm{BB}}$

TABLE SI
PARAMETERS OF THE NODAL POST-CONTINGENCY MODEL

| Parameter | Value |
|---|---|
| Nominal frequency $f_0$ | 50 Hz |
| Power base $P^B$ | 100 MVA |
| SG inertia constant $H_u$ | 3.5 s on $S_u$ |
| SG transient reactance $X_u$ | 0.5 p.u. on $S_u$ |
| TGOV1 time constants $T_1$, $T_2$, $T_3$ | 0.05, 1.0, 2.1 s |
| GFM virtual inertia $H^{\mathrm{V}}$ | 2.0 s on $S_u$ |
| GFM coupling reactance $X_u$ | 0.2 p.u. on $S_u$ |
| Droop of SGs, IBRs, and BESS units | 5% on the rating |
| IBR activation time constant $T^{\mathrm{A}}$ | 1.0 s |
| BESS response time constant | 0.1 s |
| Damping $D_u$ of SGs and GFM IBRs | 2.5 p.u. on $P^B$ |
| Damping $D_u$ of IEEE 300 SGs | 2.0 p.u. on $S_u$ |
| UFLS relay window $T_{\mathrm{r}}$ | 0.15 s |
| Simulation horizon $T_{\mathrm{s}}$ | 12 s |

TABLE SII
TRAINING SETTINGS OF THE PICNN

| Setting | Value |
|---|---|
| Hidden layers $L$, width $w$ | 3, 64 in both branches |
| Activations | ReLU (convex), ELU (context) |
| Weight initialization | $\mathcal{N}(0, 0.05^2)$, zero biases |
| Sign-constrained weights | softplus, raw values from $-2$ |
| Input and label scaling | standardized, training statistics |
| Loss | Huber, $\delta = 0.05$ (standardized) |
| Optimizer | AdamW, rate $10^{-3}$, decay $10^{-4}$ |
| Learning-rate schedule | cosine annealing, 220 epochs |
| Batch size | 256 |
| Maximum epochs | 220 |
| Early stopping | validation RMSE, patience 25 |
| Random seeds | 4, best validation RMSE kept |

and $\boldsymbol{B}_{\mathrm{BS}}$ are rebuilt for the surviving devices. The internal angles of the surviving devices are continuous, their speeds start at $\omega_u(0) = 1$ and their governor and activation states at the pre-contingency steady state, and $\boldsymbol{p}^{\mathrm{F}}(0) = \mathbf{0}$. The model is integrated over $T_{\mathrm{s}} = 12$ s with relative and absolute error tolerances of $10^{-7}$ and $10^{-9}$. For bus $n$, the relay-window drop is

$$\Delta f_{n,c}^{\mathrm{nad}}(x) = \max_{0 \le t \le T_{\mathrm{s}}} \min_{t - T_{\mathrm{r}} \le s \le t} \big[ f_0 - f_n(s) \big], \qquad \text{(S10)}$$

with $f_n(s) = f_0$ for $s < 0$ and the UFLS relay window $T_{\mathrm{r}} = 0.15$ s, and the worst-bus drop $\Delta f_c^{\mathrm{wb}}(x) = \max_{n \in \mathcal{N}} \Delta f_{n,c}^{\mathrm{nad}}(x)$ enters (1s). Table SI lists the parameters, and the unit ratings and network data are in the accompanying test-system data.

## II. PICNN TRAINING

The PICNN of Section III-B learns the worst-bus drop of the nodal model above, one label per trip at a labeled operating point. The labeled operating points of each penetration level are split at random into training, validation, and calibration sets, so that all trips of an operating point fall in the same set. On IEEE 118, these sets hold 12,528, 1,728, and 3,888 rows, and the calibration rows form the calibration sets of Section IV-A. Table SII lists the training settings. The network is trained from four random seeds, and the one with the lowest validation error is deployed. The IEEE 300 surrogate and the surrogate of the 51% fleet of Section V-E are trained with the same settings on labels of their own systems.

## REFERENCES


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